Is the Gap in the Model or the Metric? Evidence of English Bias in Korean Lip-Sync Evaluation

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***Abstract*—Generative lip-sync models such as Wav2Lip are trained and evaluated almost entirely on English, and their standard evaluation metrics come from an English-trained SyncNet. This project fine-tuned the Wav2Lip generator on a 772-clip subset of the Korean KMSAV dataset, keeping its sync expert frozen, and evaluated it on held-out Korean speakers and on English LRS3 clips over ten random seeds. Fine-tuning consistently reduced the frozen expert's validation sync loss and moved output timing closer to that of real video, yet it slightly but significantly worsened both standard metrics on Korean and lowered sync confidence on English, in every seed. The baseline itself showed no Korean-English gap under these metrics, while real, unedited Korean video scored worse than Wav2Lip's own Korean output for every test speaker and content category. Together with offset, framing and correlation analyses, these results suggest that the standard metrics favour English-like, Wav2Lip-like lip motion over faithful measurement of Korean audio-visual synchrony, and that they cannot by themselves establish whether lightweight fine-tuning closes a cross-lingual gap.**

***Index Terms*—lip-sync, Wav2Lip, SyncNet, cross-lingual evaluation, Korean, metric bias**

I. Introduction

Accurate lip-sync is central to dubbing, translated lectures and accessibility tools, but leading speaker-independent models such as Wav2Lip [1] were trained and evaluated on English data. Work extending Wav2Lip to other languages reports degraded performance [6], [7], and heavier cross-lingual architectures have been proposed [8], [9]. My proposal asked whether *lightweight*, language-specific fine-tuning of Wav2Lip's generator is enough to close this gap for Korean, a language phonetically distant from English.

The proposal already noted a risk: every comparison relies on LSE-D and LSE-C [1], which are computed by a SyncNet [2] trained on English video. During the project this risk became the main finding. Fine-tuning improved the training-time sync signal but worsened the evaluation metrics, and real Korean video scored below Wav2Lip's own Korean output. The research question therefore broadened to: *can the standard metrics measure Korean lip-sync faithfully, and what do they say about lightweight fine-tuning?*

Contributions. (1) A verified Korean preprocessing and evaluation pipeline for Wav2Lip, including parity tests against Wav2Lip's own audio code and a fixed, speaker-disjoint evaluation protocol. (2) Generator fine-tuning on KMSAV [3] with the expert frozen, repeated over 10 seeds and evaluated against the unmodified model on Korean and English using paired statistical tests. (3) Converging, though not causal, evidence that LSE metrics are biased against real Korean speech in this setting, from real-versus-generated comparisons across speakers and content categories, a training-versus-evaluation divergence, and framing, offset and correlation analyses.

Changes from the proposal. OLKAVS [5] requires Korean nationality, so KMSAV was used instead. LRS2 requires an institutional BBC licence, so English evaluation uses LRS3 [4]. The planned 30 fps adaptation was dropped (Section III-A). The phoneme-level metric stretch goal was not implemented; the results here motivate it as future work.

II. Background and Related Work

Wav2Lip [1] is the model I evaluate and fine-tune. It generates a lip-synced face frame by frame from a reference face, a lower-half-masked pose prior and a mel-spectrogram window. Its key idea is a pre-trained *lip-sync expert*, a SyncNet-style discriminator [2] that distinguishes in-sync from out-of-sync audio-video pairs and stays frozen while the generator trains, because fine-tuning it on generated faces degraded its sync accuracy. The generator minimises

$$L = (1 - s_w - s_g)\, L_{\text{recon}} + s_w\, E_{\text{sync}} + s_g\, L_{\text{gen}}, \qquad (1)$$

combining L1 pixel reconstruction, the frozen expert's sync loss $E_{\text{sync}} = -\log P_{\text{sync}}$, and an adversarial visual-quality loss. I use the variant without the visual-quality GAN ($s_g = 0$). The expert sees a fixed window of $T_v = 5$ frames (0.2 s). Wav2Lip's own ablation found that larger windows consistently improved sync discrimination, and CALS [11] showed that lip-sync quality keeps improving with up to about 1.2 s of context, which suggests Wav2Lip's short window leaves phonetic context unused.

LSE-D and LSE-C [1] evaluate sync using the separately trained SyncNet of Chung and Zisserman [2]. For each clip, SyncNet compares audio and video embeddings across offsets of ± 15 frames. LSE-D is the minimum distance (lower is better) and LSE-C is the confidence, the gap between the median and minimum distance (higher is better). Because both search across offsets, they measure how distinct the best audio-visual match is, not only whether it occurs at zero offset.

Cross-lingual lip-sync. Oskooei et al. [6] evaluated Wav2Lip on Turkish, Persian and Arabic, and Venkataraghavan et al. [7] adapted it for Indian languages. Both rely solely on LSE-D/LSE-C, a limitation because these metrics come from a SyncNet trained on English audio-visual data, so

they may carry an English-centric bias rather than measuring true phoneme-viseme correspondence in the target language. Heavier architectures have also emerged. MuEx [8] learns a phoneme-guided mixture of experts that maps between phonemes and visemes, at greater architectural cost than Wav2Lip's single generator. DiffDub [9] replaces adversarial training with a diffusion model that iteratively denoises only the lower-face region, avoiding GAN instability while leaving pose and background untouched. Phoneme- and context-aware methods such as SyncTalkFace [10] and CALS [11] likewise improve sync over Wav2Lip's short audio window.

Korean data. Tapu et al. [12] attribute the weaker performance of visual speech recognition outside English to a resource imbalance between English and other languages, and the same imbalance applies to lip-sync generation. OLKAVS [5] is the largest open Korean audio-visual corpus, but it consists of read sentences recorded in a studio. KMSAV [3] instead collects spontaneous, multi-speaker Korean conversation from Creative Commons YouTube videos across 12 content domains, with about 150 hours transcribed and per-utterance face tracks extracted with a procedure modelled on LRS3. This makes it closer in collection method to LRS3, but different in speaking style.

Gap. No prior work tests whether lightweight retraining of an English lip-sync model closes the cross-lingual gap for Korean, and none tests whether the SyncNet-based metrics that all of the above rely on measure non-English sync faithfully.

Baselines. The unmodified, pretrained Wav2Lip is the baseline, since it is the model being modified. Real, unedited test clips scored with the same pipeline provide a reference row, as in the original paper. MuEx and DiffDub cannot be compared numerically with this work: DiffDub reports LSE only on English (HDTF) and assesses Korean solely through a 12-participant user study [9], while MuEx excludes Korean from training, evaluates it only by human ranking, and does not include Wav2Lip as a baseline [8]. MuEx also defines its own landmark-based "LSE-D", which differs from the SyncNet metric of the same name used here.

III. Methodology and Implementation

A. Data pipeline

Korean. 772 manually reviewed utterance clips from 29 KMSAV videos were cropped to 224×224 face tracks using KMSAV's own bounding boxes. Each clip is spontaneous Korean speech from YouTube, with a separate 16 kHz mono audio track and a content-domain label from the dataset. The source videos are 29.97 fps, but the cropped clips are 25 fps. I kept Wav2Lip's 25 fps configuration because the frozen expert pairs $T_v = 5$ video frames with 16 mel steps (80 mel frames/s), i.e. 0.2 s, and this alignment only holds at 25 fps. The proposal's planned 30 fps change would have broken it.

English. The official LRS3 distribution has been discontinued, and the available mirrored test split contains only grayscale mouth crops, which Wav2Lip cannot use. I therefore used a fixed 100-clip subset of LRS3 trainval (the first clip from each of the first 100 videos, 25 fps, 224×224). Wav2Lip was trained on LRS2 and never saw LRS3.

Audio. I reimplemented Wav2Lip's mel recipe (80 bands, $n_{\text{fft}} = 800$, hop 200, window 800, 55–7600 Hz, pre-emphasis 0.97). During verification I found that my initial normalisation mapped to $[0, 1]$ rather than Wav2Lip's symmetric $[-4, 4]$, which would have invalidated every sync score. After fixing it, parity tests confirm my mels match those produced by Wav2Lip's own code to within 10^{-4} on Korean and English clips.

Dataset build and split. A build script writes Wav2Lip's training layout (one folder of frames plus `audio.wav` per utterance) and checks that each clip's frame count matches its audio duration at 25 fps (within ± 2 frames). All 772 clips passed. The split is *speaker-disjoint*, grouped by source video so that no speaker appears in two partitions: 597 train / 59 validation / 116 test clips, with 7 speakers in the test set. The split seed (103) was chosen by simulating 200 seeds on split proportions only, before any evaluation.

B. Verification of the sync expert and crop framing

Before training, I checked that the frozen expert scores real video correctly by comparing P_{sync} on aligned audio windows against windows shifted by 10 frames, with inputs built exactly as in Wav2Lip's training code. On an LRS3 clip, the aligned P_{sync} was only 0.174 (vs. 0.086 shifted, aligned higher in 62% of windows), because LRS3 face tracks are framed loosely: the expert sees only the lower half of a 96×96 face, so loose framing pushes the mouth out of view. Re-cropping with S3FD [15], the detector used in Wav2Lip's preprocessing, raised it to 0.789 (vs. 0.082, aligned higher in 100% of windows). On a KMSAV clip, the dataset's own crops and S3FD crops scored the same (0.502 vs. 0.509 aligned), so KMSAV is used as cropped, while LRS3 is S3FD-cropped during inference.

C. Fine-tuning

Only the generator is fine-tuned, initialised from the released `wav2lip.pth`. The expert (`lipsync_expert.pth`) stays frozen, as in the original method, so the training signal is identical to Wav2Lip's own. I adapted Wav2Lip's training script by adding: seeding (Python, NumPy, PyTorch, with DataLoader workers derived from the seed); a fresh optimiser and step counter when loading the pretrained weights; loading of my speaker-disjoint filelists; logging of validation L1 and sync loss; a fixed step budget; and fixes so the script runs on Windows (model construction moved under the main guard, since spawned DataLoader workers otherwise each loaded a model onto the GPU and ran out of memory).

D. Evaluation protocol

Following the Wav2Lip benchmark protocol, each test video is paired with audio from a randomly chosen clip of a *different speaker* that is no longer than the video (seed 0), so the model must generate new lip shapes rather than copy them from the input. Korean video is paired only with Korean audio, and English with English. The same 116 Korean and 100 English

pairs are used for every model. Generated videos are scored against the mismatched driving audio they were generated from, while the real-video reference scores each original clip against its own synchronised audio (for KMSAV, the separate utterance audio muxed back onto the clip). LSE-D, LSE-C and SyncNet's estimated audio-visual offset are computed with `syncnet_python` [2], after confirming it reproduces its reference demo output (offset 3, distance 5.352, confidence 10.087). Re-scoring the baseline outputs reproduced identical LSE values, confirming the scoring is deterministic. The minimum face-track length was lowered to 20 frames, as the default of 100 frames (4 s) discards most of these short clips. One LRS3 clip fails face detection in every condition and is excluded, giving $n = 99$.

E. Reused, modified and implemented

Reused: Wav2Lip's architecture, pretrained weights, inference code and training loop; `syncnet_python` and its weights; S3FD. **Modified:** the Wav2Lip training script (as above). **Implemented:** the audio pipeline and parity tests, the dataset build and split, the LRS3 subset extraction, the evaluation pairing, batch inference, LSE scoring with offset capture, the expert verification and offset-scan tools, checkpoint selection, and the statistical analysis.

IV. Experimental Settings

Hyperparameters. I used Wav2Lip's own settings for the no-GAN generator: Adam, learning rate 10^{-4}, batch size 16, 96×96 faces, $s_w = 0.01$, $s_g = 0$. These were taken from the original implementation, not tuned, because the research question concerns lightweight fine-tuning under the original recipe. Each run trains for 6000 steps (about 160 passes over the 597 training clips), evaluating on the validation set every 500 steps (100 batches of randomly sampled windows). The checkpoint with the lowest validation sync loss is selected, a rule fixed before any test evaluation. The test set was never used for any choice.

Repeated runs. Fine-tuning was repeated with 10 seeds (0–9). Each run took about 71 minutes on an RTX 3080 (11.8 GPU-hours for all 10), and evaluating each fine-tuned model takes about 36 minutes on CPU, so the recommended 30 runs were not practical. Seeds control data order, window sampling and worker randomness, but GPU training is not bitwise deterministic (cuDNN). The pretrained baseline is deterministic, so it was evaluated once. All conditions share the same split and evaluation pairs.

Statistics. For each language and metric, each clip's score is averaged across the 10 seeds and compared with the baseline score on the same clip and audio using a two-sided Wilcoxon signed-rank test [16], since observations are paired and score distributions need not be normal. Effect size is the matched-pairs rank-biserial correlation r. The four tests are Holm-corrected [17] at $\alpha = 0.05$. I also report 95% bootstrap confidence intervals for the mean difference (10,000 resamples of clips, RNG seed 0), and repeat the paired tests separately for each seed (Holm-corrected within each seed) to check that no conclusion rests on a single run.

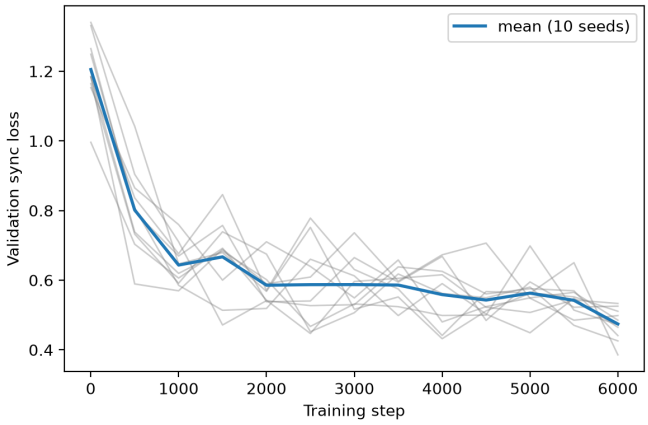


Fig. 1. Validation sync loss (frozen expert, held-out speakers) during fine-tuning. Grey: individual seeds; blue: mean of 10 seeds.

V. Results and Analysis

A. Fine-tuning reduces the training signal

Fig. 1 shows validation sync loss for all seeds. At step 1 (the pretrained model), the frozen expert gives a validation sync loss of 0.996 ($P_{\text{sync}} \approx 0.37$). Every seed improves rapidly over the first 1000–2000 steps, then plateaus with noise. The best validation sync loss across seeds is 0.386–0.485 (mean 0.451, $P_{\text{sync}} \approx 0.64$), with validation L1 also falling (0.035 to about 0.029). Best steps range from 2500 to 6000. Three seeds were still improving at step 6000, so longer training might give small further gains. By its own training objective, fine-tuning clearly works.

B. Evaluation metrics move the other way

Table I gives the main results, and Table II the paired tests. On Korean, fine-tuning *worsened* both metrics: LSE-C fell by 0.45 (95% CI $[-0.51, -0.39]$, large effect, $r = -0.92$) and LSE-D rose by 0.12 ($[+0.06, +0.19]$, small to moderate, $r = 0.36$). On English, LSE-D was unchanged ($p = 0.77$, CI includes zero), but LSE-C fell by 0.37 ($[-0.48, -0.26]$, $r = -0.62$). The LSE-C decreases are significant in every one of the 10 seeds in both languages when each seed is tested alone; the smaller Korean LSE-D increase is significant in 6 of 10. The spread across seeds (std ≤ 0.10) is small relative to these effects. Seed 9 was the most extreme run in both languages (worst on Korean, best on English), which illustrates why a single run would have been misleading.

On English, my real and baseline scores are both worse than the original paper's LRS3 results (6.956/7.592 and 6.652/7.887 [1]) by similar margins. That suggests my setup (short trainval clips, my crops, shorter minimum track length) is consistently harsher, not faulty. Absolute values are therefore not directly comparable with published numbers, but before/after comparisons on my fixed pairs remain valid.

TABLE I

LSE RESULTS. REAL/BASELINE: MEAN ± STD ACROSS CLIPS (ONE DETERMINISTIC RUN). FINE-TUNED: MEAN ± STD OF PER-SEED MEANS ACROSS 10 SEEDS.

Condition	Korean (KMSAV, $n = 116$) LSE-D ↓	LSE-C ↑	English (LRS3, $n = 99$) LSE-D ↓	LSE-C ↑
Real video	9.517±0.972	3.319±1.234	7.492±0.812	6.735±1.432
Baseline	7.781±0.837	5.898±1.254	7.704±0.897	6.057±1.491
Fine-tuned	7.901±0.101	5.449±0.067	7.686±0.040	5.689±0.080

TABLE II

PAIRED TESTS, FINE-TUNED (SEED-AVERAGED) MINUS BASELINE: MEAN DIFFERENCE WITH BOOTSTRAP 95% CI, RANK-BISERIAL r, WILCOXON p (HOLM, 4 TESTS), AND NUMBER OF SEEDS SIGNIFICANTLY WORSE WHEN TESTED INDIVIDUALLY.

Lang.	Metric	Diff. [95% CI]	r	p_{Holm}	Seeds
Korean	LSE-D ↓	$+0.119$ $[+0.06, +0.19]$	$+0.36$	1.3e-3	6/10
Korean	LSE-C ↑	-0.449 $[-0.51, -0.39]$	-0.92	3.3e-17	10/10
English	LSE-D ↓	-0.018 $[-0.13, +0.09]$	-0.03	0.77	0/10
English	LSE-C ↑	-0.368 $[-0.48, -0.26]$	-0.62	3.2e-7	10/10

C. Evidence consistent with English bias in the metrics

Three observations suggest the metrics, rather than the model, drive these results, although none is an independent measure of true synchrony.

(i) No apparent baseline gap. Under LSE, the unmodified model shows no apparent Korean-English gap (LSE-D 7.78 vs. 7.70). This does not show that no real gap exists, only that LSE does not reveal one, even though previous reports of cross-lingual degradation [6], [7] would predict one.

(ii) Real Korean video scores worst of all. Real video should be an upper reference. For English it is: real LRS3 outscores the baseline on both metrics. For Korean the order flips: real KMSAV scores far *below* Wav2Lip's own output (LSE-C 3.32 vs. 5.90, LSE-D 9.52 vs. 7.78). This is not driven by a single speaker or content type: real LSE-C is below the baseline's for all 7 test speakers and all 4 KMSAV content domains (Table III). Eating shows are not the lowest-scoring domain. The only lecture-style domain (science lectures), whose speaking style is closest to TED talks, scores highest, which suggests speaking style contributes; but even it remains below both generated output and real English, and current-events discussion scores lowest despite few speaker turn changes, so turn-taking alone does not explain the ordering. The frozen expert, also trained on English, shows the same pattern: 0.51 aligned P_{sync} on a properly framed Korean clip vs. 0.79 on English (Section III-B).

(iii) Training and evaluation diverge. Fine-tuning optimises reconstruction of real Korean lip motion alongside the expert's sync loss. If SyncNet rates real Korean lip behaviour poorly, as (ii) shows, then outputs that move towards it should score *worse*, which is what happened, in every domain (Table III). The specific pattern supports this: confidence (LSE-C) fell much more than distance (LSE-D), so SyncNet still finds a comparable best match, just a less distinct one.

TABLE III

KOREAN TEST CLIPS BY KMSAV CONTENT DOMAIN [3]: MEAN LSE-C (↑). TURNS: AVERAGE SPEAKER CHANGES PER 10 S IN EACH KMSAV DOMAIN, AS REPORTED IN [3]. EACH DOMAIN HAS ONLY 1–3 TEST SPEAKERS, SO GROUPS ARE DESCRIPTIVE ONLY.

Domain	Clips	Spk.	Turns	Real	Baseline	Fine-tuned
Beauty/makeup	13	2	3.77	3.57	5.24	4.74
Current-events discussion	27	1	0.87	2.27	5.27	4.78
Mukbang (eating show)	59	3	2.78	3.24	6.36	5.92
Science lectures	17	1	0.26	5.08	5.80	5.42

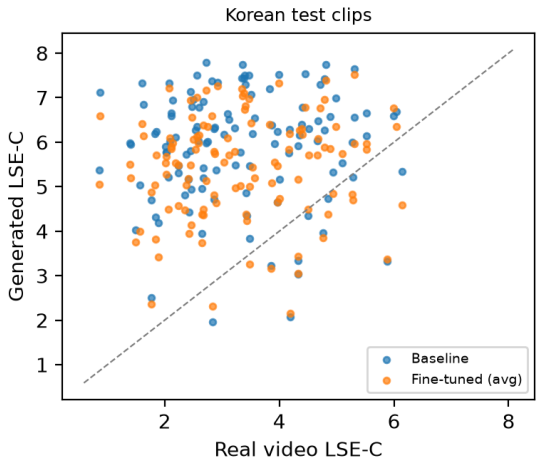


Fig. 2. Per-clip LSE-C on the Korean test set: real video (x) vs. generated output on the same frames (y). Dashed: $y = x$. Generated output scores higher for 91% of clips, and the real-video score does not predict the generated score.

Is it the video rather than the speech? If low real-video scores came from visual properties of the clips (pose, image quality, framing), the same frames should also score low when lip-synced to other audio. They do not (Fig. 2): Wav2Lip's output outscores the real video on the same frames for 91% of Korean test clips, and real-video and generated scores are essentially uncorrelated (Spearman $\rho = +0.11$ for Korean LSE-C, $p = 0.25$; $|\rho| \leq 0.19$ and $p \geq 0.056$ for every metric and language). This argues against the real-video penalty coming from how the original faces look, although generation also changes the mouth region SyncNet sees, so it does not isolate language as the cause.

Alternative explanation: timing. I checked whether KMSAV is simply out of sync, or whether fine-tuning learned poor timing from it. SyncNet's estimated offsets on real clips centre at -0.70 ± 0.48 frames for LRS3 (this pipeline's in-sync reference) and -1.16 ± 1.73 for KMSAV. An expert offset scan on one training clip from each of 10 videos found 2 aligned, 2 peaking at -1 frame, 4 at $+1$, and 2 with no measurable sync response, so KMSAV has per-video offsets of up to ± 1 frame (40 ms) and some clips with no sync signal. However, the generated outputs show that fine-tuning did not learn worse timing. Baseline Wav2Lip output sits at -2.05 (Korean) and -2.16 (English) frames, while the fine-tuned models sit at

-1.11 ± 0.06 and -1.80 ± 0.05 (mean $\pm$ std across 10 seeds). Every seed shifted about one frame *towards* real video, and on Korean the fine-tuned timing matches real KMSAV almost exactly. By SyncNet's own estimate, fine-tuning made output timing more like real speech, yet LSE-C fell. Because the offset comes from the same English-trained SyncNet, this is supporting rather than independent evidence.

D. Computational cost

Training used an RTX 3080 (10 GB) at about 4.5 steps/s. Each 6000-step run took about 71 min (range 70.6–70.9 across seeds), of which roughly 22 min is training steps; the remainder is 13 validation passes and DataLoader worker start-up at each epoch on Windows. The generator has 36.3M trainable parameters. Inference and scoring ran on an Apple M3 CPU. Per fine-tuned model, inference took about 8 min for the 116 Korean clips and 12 min for the 99 English clips (which also run S3FD face detection), and LSE scoring took about 9 and 7 min respectively. Training and evaluation used different hardware, so no timing comparisons are drawn across them.

VI. Discussion

Supported by the experiments. Lightweight generator fine-tuning reliably improves the frozen expert's sync loss on held-out Korean speakers and moves output timing towards real video, but under LSE it slightly and significantly degrades Korean sync and lowers English sync confidence, in every seed. Under LSE, the baseline shows no cross-lingual gap, and real Korean video scores below generated video for every test speaker and content domain, the reverse of English. The real-video penalty is not explained by visual clip properties, eating content, or timing offsets.

Plausible but not demonstrated. That the cause is English bias in SyncNet specifically. Language is still partly confounded with speaking style: KMSAV is mostly spontaneous multi-speaker conversation, while LRS3 is rehearsed TED talks. The lecture-style Korean domain scoring highest suggests speaking style matters, but with one speaker per domain this could also be an individual-speaker effect, and a gap to English remains even there. A Korean-trained SyncNet, or rehearsed Korean lectures from many speakers, would be needed to separate language from style. The English LSE-C drop could reflect forgetting of English-specific lip motion, the shift towards KMSAV-like timing, or both.

Implication. Cross-lingual lip-sync studies that report only LSE [6], [7] may under- or over-state gains for non-English languages. In this setting, a model that moves towards real Korean speech appears to be *penalised* by the metric. This is consistent with prior cross-lingual work: DiffDub reports that Wav2Lip outscores it on LSE despite being rated worse by human evaluators, attributing this to Wav2Lip directly optimising the metric [9], and both DiffDub and MuEx rely on human evaluation rather than SyncNet-based metrics for their Korean results [8], [9]. The original question, whether lightweight fine-tuning closes the cross-lingual gap, cannot be answered with LSE alone.

Limitations. The 116 Korean test clips come from only 7 speakers (videos), and content domains each contain only 1–3 speakers, so domain and speaker effects cannot be separated. The 100 English clips come from LRS3 trainval rather than the official test set. There was no independent measure of synchrony (human evaluation or a Korean-trained SyncNet), so LSE changes cannot be equated with changes in perceived sync quality, and there was no hyperparameter search. The validation signal is noisy (59 clips, random windows), and 3 of 10 runs had not plateaued by step 6000. KMSAV timing offsets were measured but not corrected.

VII. Ethical Considerations

Misuse. Lip-sync models can be used to create deceptive video. Following the Wav2Lip authors' guidance, all generated outputs are treated as synthetic, and no generated video of identifiable people is published. The repository releases code and aggregate metrics only. **Data and consent.** KMSAV (CC BY-NC-SA 4.0) contains public YouTube creators who did not consent to lip-sync research specifically; it is used non-commercially and not redistributed. LRS3 is CC BY 4.0, with TED's non-commercial, no-derivatives terms on the underlying talks; no LRS3 media, original or generated, is redistributed. **Bias and representation.** This project's central finding is itself an equity concern: English-trained evaluation tools can misjudge systems for other languages, which could lead to worse dubbing and accessibility tools for non-English speakers being reported as adequate, or good ones as failures. **Compute.** Ten fine-tuning runs used about 12 GPU-hours on a single consumer GPU, plus CPU time for evaluation; the low cost is part of what makes lightweight fine-tuning attractive, but it is still a cost. **Deployment.** This is a classroom research prototype evaluated on a small dataset with automatic metrics only, and it is not suitable for production dubbing.

VIII. Conclusion and Future Work

I built and verified a Korean evaluation and fine-tuning pipeline for Wav2Lip and fine-tuned its generator over 10 seeds. Fine-tuning consistently improved Wav2Lip's own training-time sync signal and moved output timing towards real video, but slightly worsened LSE on Korean and reduced sync confidence on English in every seed, while real Korean video scored below generated video for every test speaker and domain. My answer to the research question is therefore that the standard metrics cannot establish whether lightweight fine-tuning closes the gap, as they appear to favour English-like, Wav2Lip-like lip motion over Korean synchrony. The main limitations are the few test speakers, the language–style confound and the lack of human evaluation. Next steps are: (1) the proposed phoneme-level diagnostic, using the Montreal Forced Aligner [14] or zero-shot phoneme recognition [13]; (2) a Korean-calibrated SyncNet; (3) a small human perceptual study; and (4) per-video offset correction and non-speech filtering.

STATEMENT

I, James Brooker, state that the research, experiments and conclusions in this report are my own work. Tools used: Python, PyTorch [18], librosa [19], NumPy/SciPy/pandas, ffmpeg, Wav2Lip's code and pretrained models [1], `syncnet_python` [2], S3FD [15], PyCharm, Git/GitHub and LaTeX. Generative AI (Claude, Anthropic) was used throughout the project: for debugging, drafting graph drawing scripts and some pipeline scripts, and for revising/grading parts of this report's text, which I reviewed, verified against my results, and revised. Parts of the report have been gramatically and structurally edited by AI. All experiments were run by me, and all reported numbers come from the result files in my repository. Datasets: KMSAV [3] and LRS3 [4]. Code:Click Here

REFERENCES

[1] K. R. Prajwal, R. Mukhopadhyay, V. P. Namboodiri, and C. V. Jawahar, "A lip sync expert is all you need for speech to lip generation in the wild," in *Proc. 28th ACM Int. Conf. Multimedia (MM '20)*, Seattle, WA, USA, 2020, pp. 484–492, doi: 10.1145/3394171.3413532.

[2] J. S. Chung and A. Zisserman, "Out of time: Automated lip sync in the wild," in *Computer Vision – ACCV 2016 Workshops*, Springer, 2017, pp. 251–263.

[3] K. Park, C. Oh, and S. Dong, "KMSAV: Korean multi-speaker spontaneous audiovisual dataset," *ETRI Journal*, vol. 46, no. 1, pp. 71–81, 2024, doi: 10.4218/etrij.2023-0352.

[4] T. Afouras, J. S. Chung, and A. Zisserman, "LRS3-TED: A large-scale dataset for visual speech recognition," arXiv:1809.00496, 2018.

[5] J. Park, J.-W. Hwang, K. Choi, S.-H. Lee, J. H. Ahn, R.-H. Park, and H.-M. Park, "OLKAVS: An open large-scale Korean audio-visual speech dataset," arXiv:2301.06375, 2023.

[6] A. Rafiei Oskooei, E. Yahsi, M. Sungur, and M. S. Aktas, "Can one model fit all? An exploration of Wav2Lip's lip-syncing generalizability across culturally distinct languages," in *Computational Science and Its Applications – ICCSA 2024 Workshops*, Springer, 2024, pp. 149–164, doi: 10.1007/978-3-031-65282-0_10.

[7] V. Venkataraghavan, S. Sivapatham, and A. Kar, "Wav2Lip bridges communication gap: Automating lip sync and language translation for Indian languages," *IEEE Access*, vol. 13, pp. 72906–72917, 2025, doi: 10.1109/ACCESS.2025.3562883.

[8] Z. Su, K. Wei, J. Li, J. Kong, X. Yang, and C. Deng, "A bridge from audio to video: Phoneme-viseme alignment allows every face to speak multiple languages," arXiv:2510.06612v2, 2026.

[9] T. Liu, C. Du, S. Fan, F. Chen, and K. Yu, "DiffDub: Person-generic visual dubbing using inpainting renderer with diffusion auto-encoder," in *Proc. IEEE ICASSP*, Seoul, South Korea, 2024.

[10] S. J. Park, M. Kim, J. Hong, J. Choi, and Y. M. Ro, "SyncTalkFace: Talking face generation with precise lip-syncing via audio-lip memory," in *Proc. AAAI*, vol. 36, 2022, pp. 2062–2070.

[11] S. J. Park, M. Kim, J. Choi, and Y. M. Ro, "Exploring phonetic context-aware lip-sync for talking face generation," arXiv:2305.19556, 2023.

[12] R. Tapu, B. Mocanu, and I.-C. Chiva, "Multimodal visual speech recognition for under-resource languages via cross-modal learning and large language models," *Rom. J. Inf. Sci. Technol.*, vol. 29, no. 1, pp. 53–64, 2026, doi: 10.59277/ROMJIST.2026.1.05.

[13] Q. Xu, A. Baevski, and M. Auli, "Simple and effective zero-shot cross-lingual phoneme recognition," arXiv:2109.11680, 2021.

[14] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, "Montreal Forced Aligner: Trainable text-speech alignment using Kaldi," in *Proc. Interspeech*, 2017, pp. 498–502.

[15] S. Zhang, X. Zhu, Z. Lei, H. Shi, X. Wang, and S. Z. Li, "S^3FD: Single shot scale-invariant face detector," in *Proc. IEEE ICCV*, 2017, pp. 192–201.

[16] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[17] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[18] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[19] B. McFee *et al.*, "librosa: Audio and music signal analysis in Python," in *Proc. 14th Python in Science Conf.*, 2015, pp. 18–25.